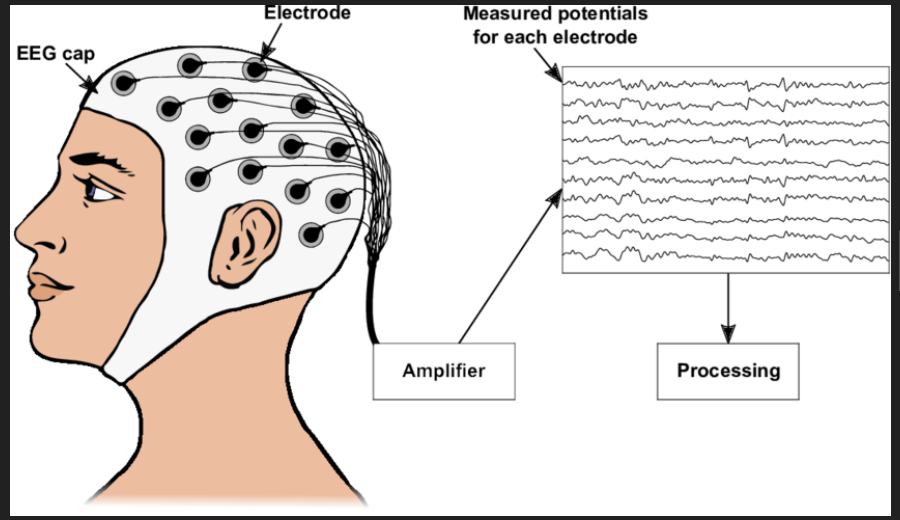
EEG Signal Classification CPEN 355 - UBC 2024

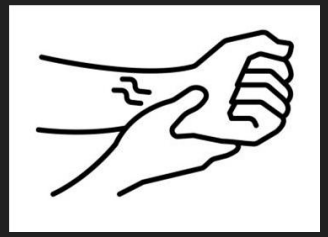
Group 8 - Adin Mauer , Sikher Sinha , Weifeng Ke

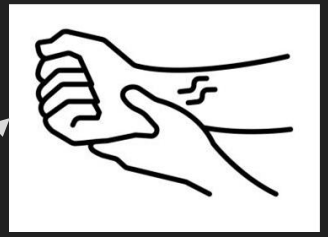
EEG Classification problem

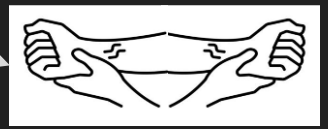
<u>Input:</u> multi-channel EEG recording

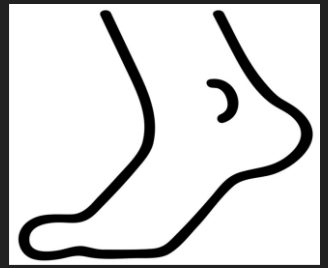


We know….
Unreliable network …





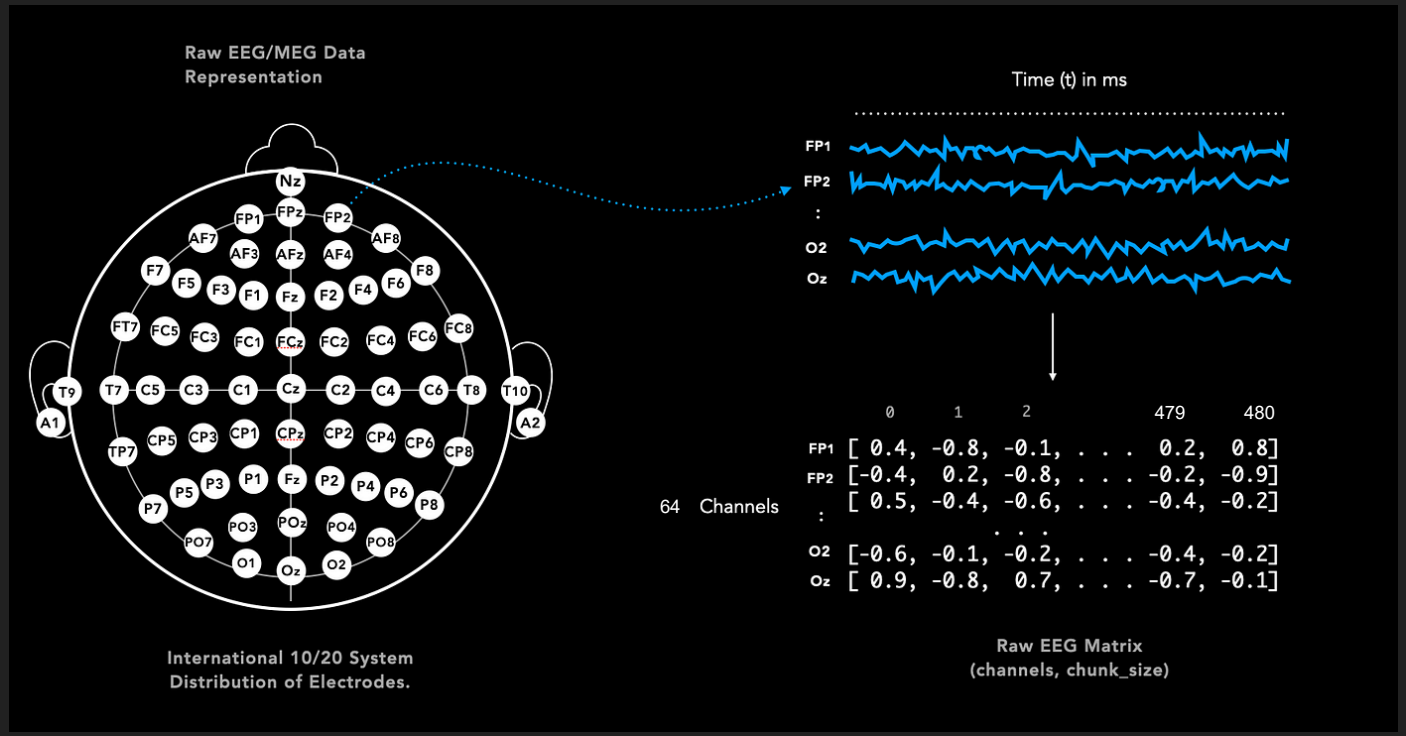




<u>Output:</u> 1 of 4 possible motor-movements

Data Preprocessing

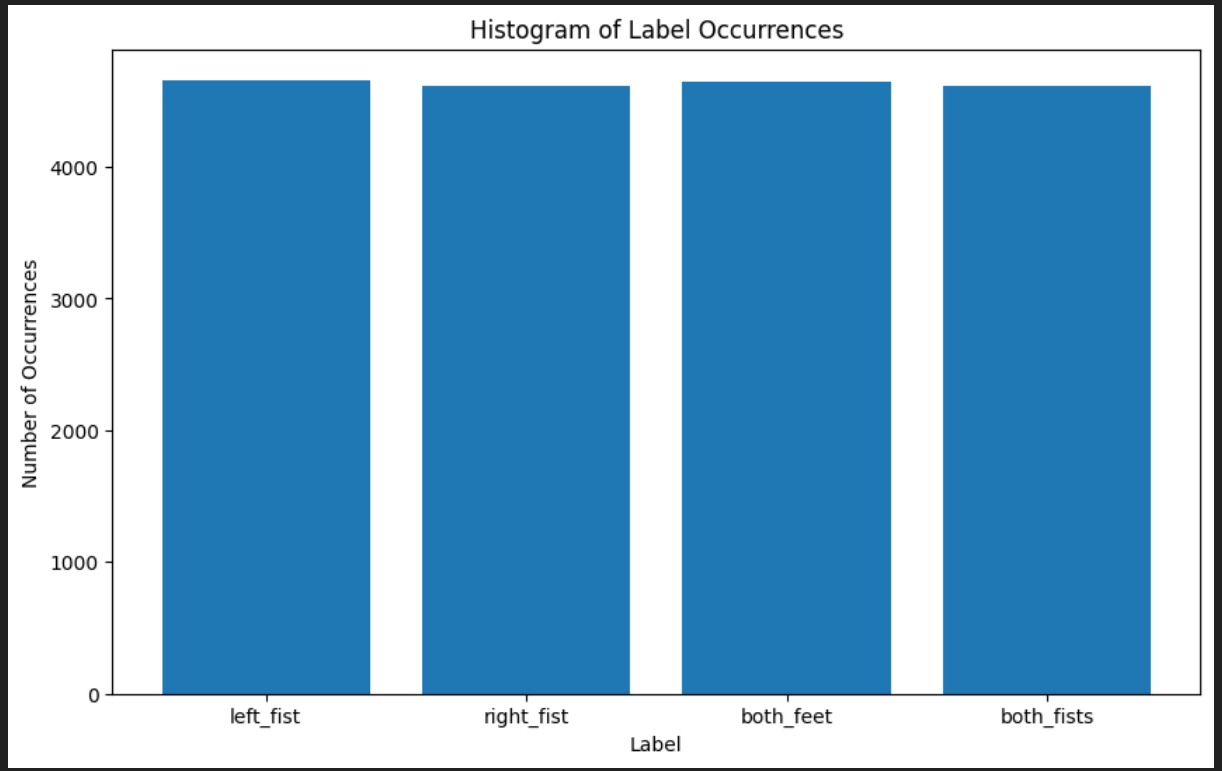
1. Download data using git annex
2. Build dataset structure



Dataset Histogram

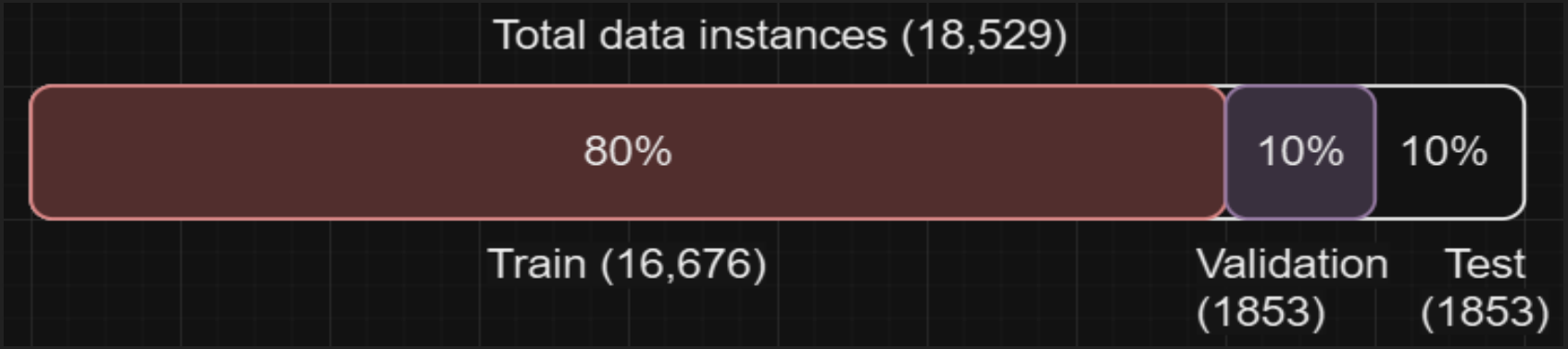
There are 18,529 labeled event instances (inputs).

Dataset is balanced (with +/- 5%)

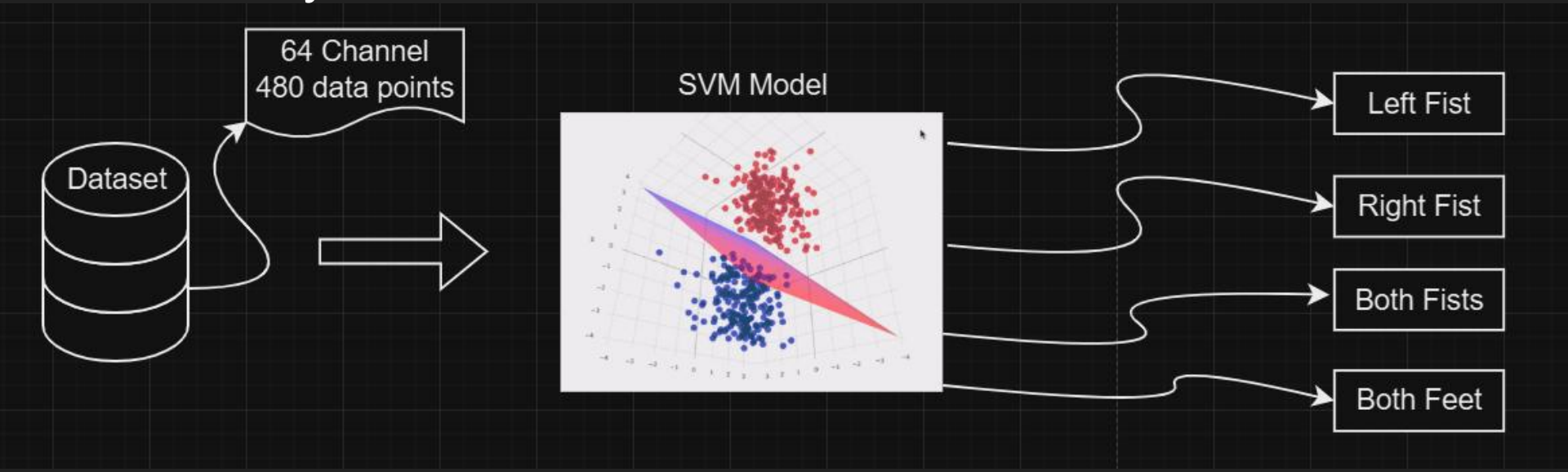


Data Preprocessing

1. …
2. …
3. Save dataset to file (2.2 GB)
4. Load dataset file
5. Dataset split



Secondary Model - SVM



Random Search Hyper Parameters:

Hyper Parameters={‘C’: 0.373, ‘gamma’: 0.121, ‘kernel’: sigmoid}

Best Validation Accuracy: 0.518

Primary Model - CNN

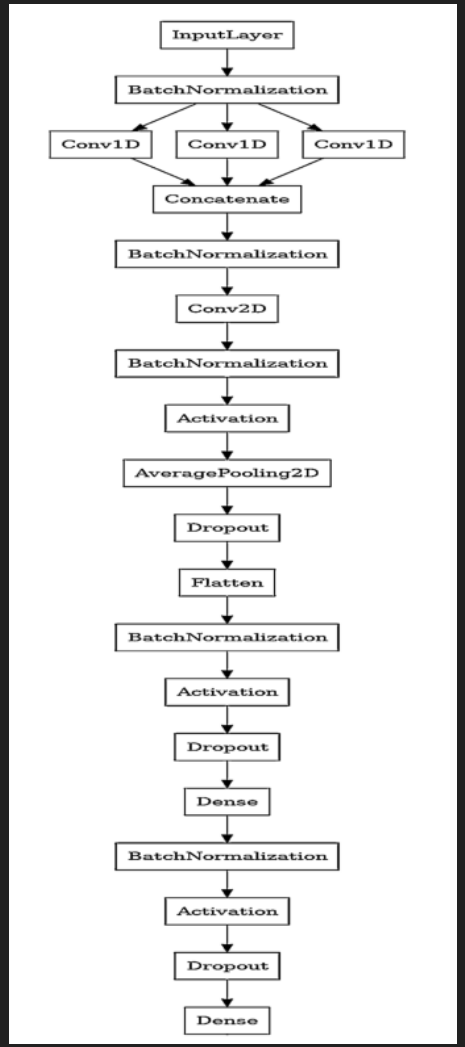
CNN captures both **temporal (time-domain** and **spatial** (inter-channel) relationships and features.

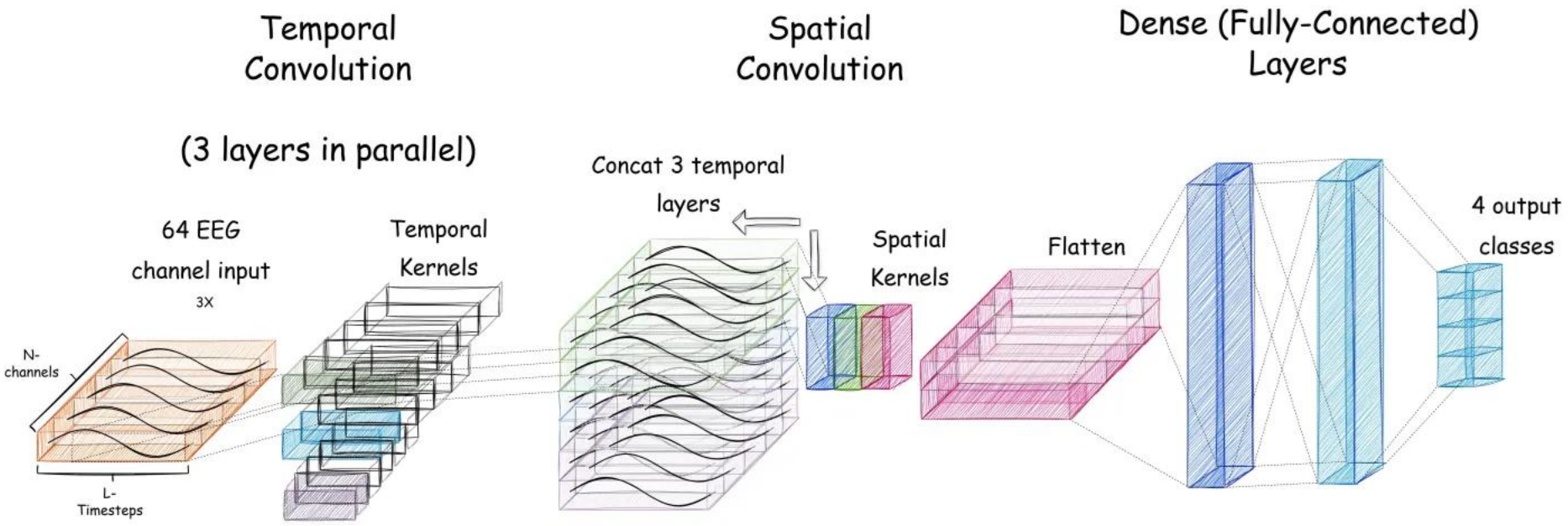
<u>Optimizer:</u> Adam (SGD - momentum)

<u>Activation:</u> ReLU

<u>Cost/Loss Function:</u> Cross-entropy

<u>Additional Details:</u> Includes **dropout** regularization. Includes batch normalization and **average pooling** layers.





* Does not show batch-normalization, dropout or average pooling layers

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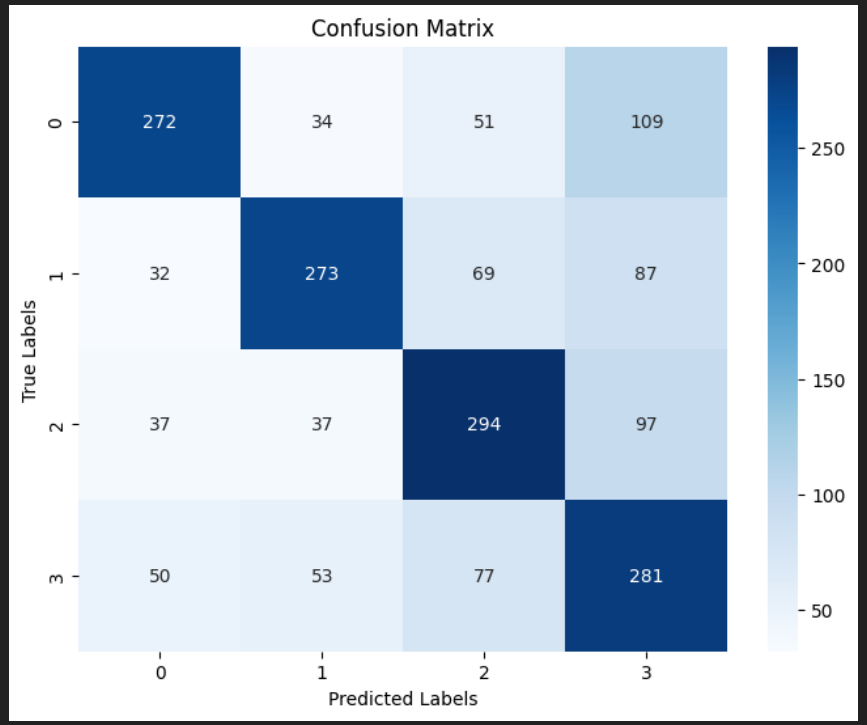
CNN VS SVM Test Performance

Accuracy of CNN: 61.03%

CNN	Class 0:	Class 1:	Class 2:	Class 3:	Average
Precision	0.7	0.68	0.59	0.51	62%
Recall	0.57	0.62	0.62	0.63	58.5%
F1-Score	0.63	0.65	0.61	0.57	61.5%

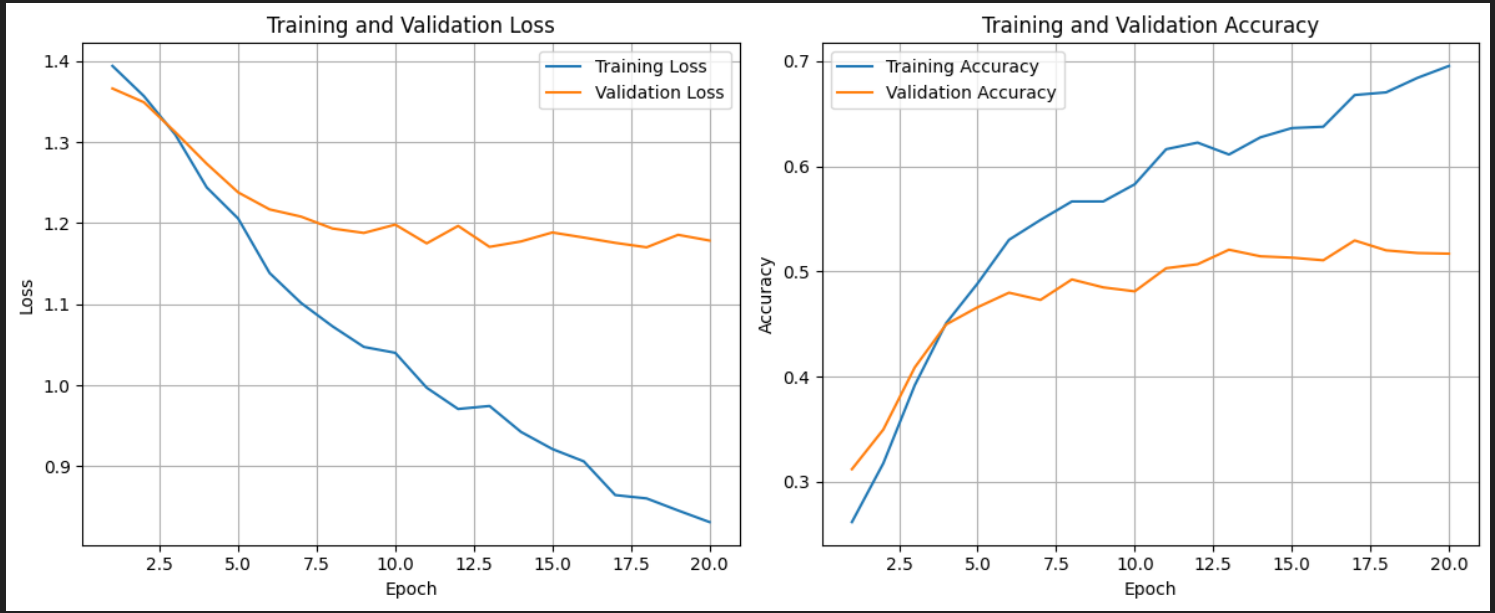
Accuracy of SVM: 56.2%

SVM	Class 0:	Class 1:	Class 2:	Class 3:	Average
Precision	0.62	0.81	0.7	0.40	63%
Recall	0.6	0.24	0.71	0.69	56%
F1-Score	0.61	0.38	0.71	0.51	55%



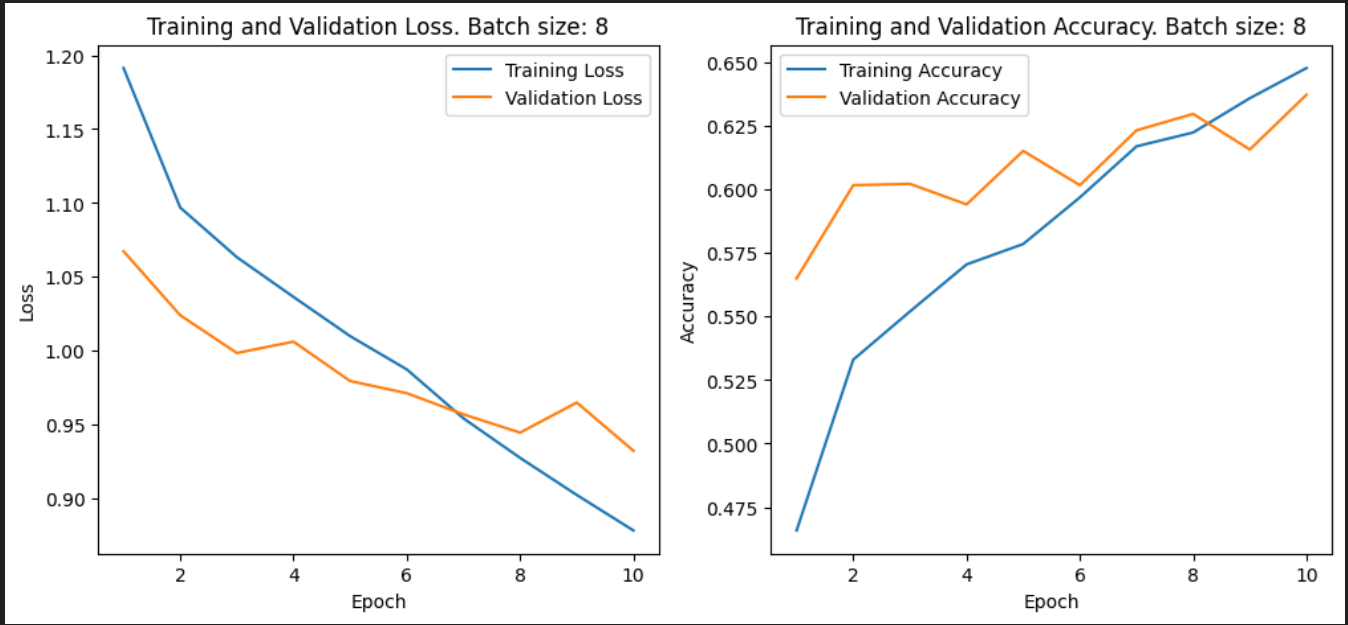
Accuracy, loss curve VS epochs observation

Overfitting over time - too many epochs?



Accuracy, loss curve VS epochs observation

Room to improve - need more epochs?
Also - maybe unstable? Slower learning rate?



Questions?